

The impact of dietary protein on calcium absorption and kinetic measures of bone turnover in women.



Although high-protein diets induce hypercalciuria in humans, the source of the additional urinary calcium remains unclear. One hypothesis is that the high endogenous acid load of a high-protein diet is partially buffered by bone, leading to increased skeletal resorption and hypercalciuria. We used dual stable calcium isotopes to quantify the effect of a high-protein diet on calcium kinetics in women. The study consisted of 2 wk of a lead-in, well-balanced diet followed by 10 d of an experimental diet containing either moderate (1.0 g/kg) or high (2.1 g/kg) protein. Thirteen healthy women received both levels of protein in random order. Intestinal calcium absorption increased during the high-protein diet in comparison with the moderate (26.2 $\pm$ 1.9% vs. 18.5 $\pm$ 1.6%, $P < 0.0001$, mean $\pm$ sem) as did urinary calcium (5.23 $\pm$ 0.37 vs. 3.57 $\pm$ 0.35 mmol/d, $P < 0.0001$, mean $\pm$ sem). The high-protein diet caused a significant reduction in the fraction of urinary calcium of bone origin and a nonsignificant trend toward a reduction in the rate of bone turnover. There were no protein-induced effects on net bone balance. These data directly demonstrate that, at least in the short term, high-protein diets are not detrimental to bone.